Abstract

Traffic accidents continue to be a leading cause of deaths and injuries worldwide, with limited time for human drivers or autonomous systems to respond. There has been significant progress in accident prediction using deep learning but most models remain black-box in nature, lacking explainability, making them difficult to interpret and even harder to trust in safety critical environments. This research develops a structurally explainable framework that anticipates traffic accidents from egocentric dash-cam video through hierarchical probabilistic modelling. We represent accident risk as a multi-level latent structure where observable driving scene features influence intermediate risk factors (Collision Imminence, Behavioural Risk, and Environmental Hazard) which collectively determine top-level Accident Risk states (Safe, Elevated, Critical). Object detection and tracking are performed using YOLOv11 with DeepSORT and ego-motion compensation to extract temporally consistent safety relevant features. We employ Dynamic Bayesian Networks with the DYNOTEARS framework to learn the hierarchical dependency structure among observable features, intermediate latent factors, and accident risk across time. Given the learned structure, we perform approximate inference using Loopy Belief Propagation and variational methods to estimate real-time risk state probabilities. This hierarchical latent variable approach enables interpretable explanations that trace risk propagation through multiple levels of abstraction providing transparency that stakeholders can validate against traffic safety domain knowledge. By explicitly modelling the pathway from observations to risk through interpretable intermediate factors rather than end-to-end black-box prediction, this research offers a robust and explainable alternative to black-box deep learning models, contributing to safer and more accountable vehicle intelligence systems.

Keywords: accident prediction; explainable AI; Egocentric video.

Contents

1 Introduction **2**

2 Background **4**

3 Related Work **5**

3.1 Object Detection & tracking 5

3.2 Spatio-Temporal Relationship Modelling 6

3.3 Explainability in Accident Anticipation 7

4 Methodology **9**

4.1 Problem Statement 9

4.1.1 Assumptions 10

4.1.2 Inference 11

4.2 Purpose Statement 11

4.3 Research Questions and Experiments 11

4.3.1 Dataset 14

4.3.2 semi supervised annotation 15

References **17**

Chapter 1

Introduction

In fast-moving traffic environments, the difference between a near-miss and a fatal collision is often measured in seconds. While modern vehicles are increasingly equipped with sensors and AI-driven systems, most traffic accidents still occur without warning, leaving little room for human or machine intervention. According to the World Health Organisation [2023], over 1.19 million people die annually from road traffic accidents and between 20-50 million are left with non-fatal injuries, a staggering reminder that despite progress in vehicle automation and perception, the ability to anticipate dangerous situations before they unfold remains an unsolved and urgent challenge.

An increasing number of drivers are equipping their vehicles with dash-camera technology as it has become more easily accessible and beneficial on the road. In this paper we will focus on Egocentric visual data which is video data captured with the first person view of road traffic; this can be either by a dash-camera mounted on the car or a cellphone or any device that can give us a first person view of traffic. Figure 1.1 below shows an image with an egocentric view of traffic. Video footage from dash cameras can be used in various ways, we will use this data to anticipate possible vehicle collisions and alert the driver or to take the appropriate action in the case of an autonomous vehicle.

A key limitation in current accident prediction systems lies in their inability to reason transparently and reliably under real-world uncertainty. Many current approaches rely on deep learning using neural networks applied to multiple sensory data, which can detect patterns but often function as black boxes with limited interpretability. This presents a critical problem in safety-critical domains like autonomous driving, where accountability, trust and decision justification are essential. Existing models also typically lack structured representations of how visual, contextual and latent factors interact over space and time to influence accident risk. This presents a need

for predictive frameworks that can model the spatio-temporal dependencies explicitly while remaining interpretable to the various stakeholders.

This research aims to address this need by investigating how structured, explainable models can be used to anticipate accidents in egocentric driving scenes to support causal, transparent and trustworthy reasoning to provide value to human stakeholders or justify decisions made by autonomous driving systems.

The remainder of this proposal is organised to address the research objectives and validate the proposed contributions. Chapter 2 is the background which introduces certain concepts necessary for understanding the rest of the paper. Chapter 3 discusses the related work, reviewing the existing literature and identifying the gap this research aims to fill. Chapter 4 formally presents the technical problem statement and the research questions which we will explore and evaluate to fully address the problem. This chapter also describes the data and outlines the data preparation, exploration and evaluation metrics that will be employed. Chapter 5 presents a research plan which includes the timeline in which research milestones are expected to be met as well as potential risks associated with this research.

Figure 1.1: Dash-cam traffic footage

Chapter 2

Background

Here we briefly introduce concepts and models necessary to understand the workings of this research, especially for readers without the technical background that would make this paper easier to understand.

Yolo architecture - The Yolo (You only look Once) is a family of deep learning models specifically designed for real-time Object detection, with a focus on speed,accuracy and efficiency. Nikhileswara Rao [2024], and Hidayatullah *et al.* [2025] does a deep comparison of the different Yolo architectures and their performances and we can see from this study that the YoloV11 architecture is most adaptive for our needs as performance has improved and it is also shown an improvement on detecting smaller objects in a frame. We see the YoloV11 architecture in 2.1 as per Nikhileswara Rao [2024].

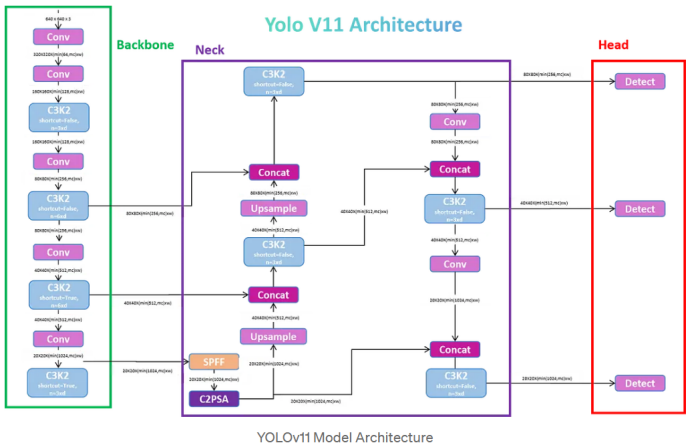

Figure 2.1: Architecture of YoloV11

Chapter 3

Related Work

In this section we review the related literature focusing on early accident anticipation using video-data. We group the reviewed literature in themes according to the important components that build up our proposed framework. We focus on key contributions in object-detection and tracking, spatio-temporal modelling for risk prediction and review contributions in explainable AI for accident anticipation. Emphasis is placed on methods that integrate computer vision with interpretable reasoning frameworks, outlining their contributions and highlighting key limitations within the context of real-world driving scenarios. This review serves as the foundation and rationale behind our proposed research.

3.1 Object Detection & tracking

To form a good foundation for an accident anticipation framework we need a highly accurate and efficient object detection and tracking pipeline suited for real-time problems. In recent years the development of Deep Learning and GPU computing power has seen the field of object detection and tracking make good progress. Object detection methods can be classified as either one stage or two stage detectors. We focus on reviewing one stage detectors because they are more efficient, as speed is critical in our domain, even though they are less accurate than two stage detectors. Wang *et al.* [2018] proposed a novel real-time object detection system combining PeeleNet with Single Shot Detection (SSD) using a lightweight backbone applied on a mobile device was crucial due to limited resources on the device, this architecture was optimized for fast processing speed. Fu *et al.* [2017] proposes a Deconvolutional-SSD (DSSD) which is an improvement to the SSD object detector by introducing additional context to their framework by using a deconvolution layer. Sairung *et al.* [2023] and Yashina *et al.* [2025] employed deep learning-based YOLO architectures for object detection from

video data across varying traffic environments.Yashina *et al.* [2025] utilized YOLOv11 for real-time object detection using a static bird's-eye view camera at a busy intersection to analyse traffic dynamics and improve flow control. In contrast, Sairung *et al.* [2023] applied YOLOv7 on dash-cam video data to extract objects using bounding boxes, which were then used to predict the probability of collisions via logistic regression. This shows that object detectors are a vital part of solving vision problems and their different architectures are adapted for different problems.

YOLO architecture demonstrates strong computational performance and detection accuracy in both static and dynamic contexts. Hidayatullah *et al.* [2025] provides a comparative analysis of different YOLO versions, highlighting the trade-offs between speed, accuracy, and robustness. In this research, we adopt YOLOv11 due to its high computational efficiency, detection precision, and robustness in dynamic traffic scenes.

3.2 Spatio-Temporal Relationship Modelling

After Detecting and tracking the objects we need a way to model the relationships between the variables in a frame (spatial) and how these evolve over time (temporal). Recent literature has leveraged deep neural networks for modelling spatio-temporal relationships to tackle challenges across various domains. Different variations of CNN-RNN combinations to have been used to implicitly model the spatio-temporal relationships in time-series data with spatial dependencies, some examples include Delasalles *et al.* [2019], Shi *et al.* [2015] and Yi *et al.* [2023]. With Yi *et al.* [2023] introducing an attention mechanism to improve the deep learning architecture. The CNN-RNN combination has mainly been used for tasks such as forecasting and imputation but this deep learning framework cannot be used for inference or discovering causality because the relationships between the variables are not explicitly modelled, they are implicitly modelled in the deep learning architecture. We aim to explicitly model these relationships through a structural representation of spatio-temporal relations, that enables interpretable risk inference and temporal prediction.

For structured representations we consider graph based approaches for spatio-temporal relationship learning, we have Delasalles *et al.* [2019] who proposed combining graph-based spatial dependency modelling with recurrent sequence learning to formulate a Spatio-Temporal graph Neural Network which showed good predictive performance over datasets in various fields.Bao *et al.* [2020] and Yu *et al.* [2018] use a Graph Convolutional Network (GCN) to capture the spatial interactions among objects and a recurrent neural network for learning temporal dependencies and latent patterns over time. Graph representations have shown to improve the learning and training speed with fewer parameters and effectively capture comprehensive spatio-temporal

relations. While capable of learning complex spatio-temporal patterns, Graph Convolutional Networks, rely on continuous latent representations that lack domain-grounded interpretability. The embeddings produced through graph convolutions do not correspond to intuitive concepts that experts can directly validate. Our approach addresses this limitation by employing structured probabilistic modelling where spatio-temporal dynamics are represented through interpretable discrete states and explicit transition probabilities, providing model-intrinsic explainability rather than requiring post-hoc interpretation methods.

3.3 Explainability in Accident Anticipation

For a model to be applicable in the field of accident anticipation, it is important that the model has a good level of explainability to foster trust. The advancements in the performance of deep neural network models for prediction problems has led to a wide adoption of these methods for research in accident anticipation but it remains difficult for these methods to be adopted in practice because of their black-box nature. Selvaraju *et al.* [2017] and Karim *et al.* [2022] proposed a post-hoc explainability method for a wide range of Convolutional Neural Networks which is called Grad-CAM (Gradient-weighted Class Activation Mapping), this provides visual explanations in the form of saliency maps to show which areas of the image or video the convolutional layers focused on to make a specific prediction. The Grad-CAM improves trust for CNN based deep learning models in this field and can be useful for debugging the model to improve performance but it does not provide any structural insight. Such post hoc deep learning methods rely heavily on deep neural networks which by nature is a black box. In addition deep neural networks can be overconfident in false predictions, leading to higher risks of mistakes caused by the system, Bao *et al.* [2020] introduced a predictive uncertainty based approach using Bayesian Networks as a layer of explainability to show the confidence level of the model in making the prediction. In contrast to the post-hoc methods we propose a method that does not rely entirely on deep neural networks but introduces structural explainability by explicitly modelling the spatio-temporal relations to enable risk inference. While our model learns predictive associations, its interpretable structure allows domain experts to reason about potential causal mechanisms underlying the accident risk. The results of the Related work can be seen in the table 3.1 below, we compare our performance results with the work listed below.

While prior work has advanced in object detection and explainability in Car Accident anticipation, there is little work on integrated probabilistic frameworks that supports both structured inference and traceable decision paths. This forms the core research

Table 3.1: Related Work Results Comparison

Paper	Method	Dataset	performance
Yi *et al.* [2023]	Dynamic Spatial-temporal attention network	AVA	F1 score=0.614 & Accuracy = 0.630
Bao *et al.* [2020]	CNN + RNN + BNN	DAD, A3D & CCD	TTA = 4.74 , AP = 99.5, mAU = 0.0137 ,mEU =0.0001
Karim *et al.* [2022]	GRU Network with Grad-CAM Saliency map	CCD	AP=93.77 & mTTA = 4.45s

contribution this paper seeks to address.

Chapter 4

Methodology

4.1 Problem Statement

This research focuses on modelling the progression of dynamic traffic scenes over time. Modelling this temporal progression requires a compact representation that captures dependencies between observed objects, environmental factors and other accident risk indicators in the traffic scene as it unfolds over time.

To structure the relationship between the variables, we employ a template-based probabilistic framework that compactly represents the system's temporal dynamics. Structuring the problem in this way will enable us to determine the likelihood of future outcomes by capturing dependencies between key variables such as object motion, environmental conditions and latent variables at each time step.

At each time step t, the system is represented as a state $\mathcal{X}^t$, which is a set of observed variables and hidden variables in the driving scene. We have

$$\mathcal{X}^t = \{\mathbf{X}^{(t)}, \mathbf{E}^{(t)}, \mathbf{O}^{(t)}, \mathbf{Y}^{(t)}\}$$

where $\mathbf{O}^{(t)}$ is the set representing all observable objects in the scene at time t (i.e cars, pedestrians, traffic lights) and $\mathbf{E}^{(t)}$ is the set representing the environmental variables that affect our observations e.g motion blur, weather conditions, lighting, etc. $\mathbf{X}^{(t)}$ is the set of high-level context features e.g relative speeds,distances,lane positions. $\mathbf{Y}^{(t)}$ is the accident risk indicator (probability of a collision in the next few seconds), modelled as a latent variable. For instance if a car swerves in wet weather and a pedestrian enters a cross-walk, these variables are observed and through their interactions over t, we can infer the state of $\mathbf{Y}^{(t)}$.

4.1.1 Assumptions

Assumption 1: Time Granularity

We assume that discrete time intervals can adequately represent our system. The driving scene can be represented by video frames, sampled at a fixed intervals (Δ) (i.e 10 FPS). We represent $\mathcal{X}^{(0:T)}$ ={$\mathcal{X}^{(0)}$, $\mathcal{X}^{(1)}$,..., $\mathcal{X}^{(T)}$ }. Where $\mathcal{X}^{(t)}$ represents a snapshot of the system state at time t. This simplifies the modelling of motion trajectories and allows us to associate variables such as objects, spatial relations and accident risk with frame-aligned measurements. From this assumption, the joint probability distribution

$$P(\mathcal{X}^{(0:T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(0:t)}) \tag{4.1}$$

This discretization allows us to build a model where temporal transitions are captured as probabilistic dependencies.

Assumption 2: Markov Assumption

Our second simplifying assumption of the system state is a first-order Markov process. The future state depends only on the present state and not on the full history. This simplifying assumption reduces model complexity and provides us with a framework to simplify our probability distribution to the following:

$$P(\mathcal{X}^{(0:T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(t)}) \tag{4.2}$$

This assumption also reflects the realistic short-term interactions in driving scenes(e.g how a sudden deceleration at t affects risk at $t+1$ without requiring long-term memory). This assumption also means that the historical information embedded at time t is enough to infer the system state at $t+1$.

Assumption 3: Time invariance assumption

We assume our system is time invariant. This allows us to learn a generalizable transition model. Formally for any $t \geq 0$, we assume: $P(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(t)})$ is the same for all t. We can therefore represent our distribution as a transition model $P(\mathcal{X}' \mid \mathcal{X})$, so that for any $t \geq 0$:

$$P(\mathcal{X}^{(t+1)} = \xi' \mid \mathcal{X}^{(t)} = \xi) = P(\mathcal{X}' = \xi' \mid \mathcal{X} = \xi) \tag{4.3}$$

This allows us to learn a single general structure and parameter set instead of finding a new model at each time step. These assumptions, while simplifying, retain the

expressive power needed to model dynamic, uncertain driving environments.

4.1.2 Inference

A central task in probabilistic inference is computing the probability of a set of events conditioned on a given set of observations called evidence. This involves reasoning over potentially high-dimensional joint distributions. The primary difficulty in probabilistic inference comes from the exponential growth in the number of variables n in the joint distribution. This complexity can be expressed as $O(2^n)$, making direct computation impractical for models with even a moderate number of variables, thus classifying this problem as NP-Hard.

In our domain we let $\mathbf{Y}^{(t)}$ represent the hidden accident risk state at time t, a latent variable that cannot be directly observed but must be inferred from observable features in the system state $\mathcal{X}^{(t)}$. Given the above assumptions, the goal is to estimate the most probable outcome of the accident variable $\mathbf{Y}^{(t)}$ conditioned on the current and prior observations which corresponds to computing the Maximum A Posteriori (MAP) estimate given by:

$$\mathrm{MAP}(Y^t = y \mid \mathcal{X}^{(t-1)}, \{\mathcal{X}^{(t)} - Y^{(t)}\}) = \arg\max_x P(Y^t \mid \mathcal{X}^{(t-1)}, \{\mathcal{X}^{(t)} - Y^{(t)}\})$$

4.2 Purpose Statement

The purpose of this research is to investigate how accident risk can be modelled as a latent temporal process and inferred from observable spatio-temporal features in egocentric driving scenes using probabilistic temporal models for early accident anticipation. By formulating accident risk as a hidden variable that evolves over time, we model the progression through interpretable discrete states. We aim to evaluate the accuracy, interpretability, and computational feasibility of structure learning and inference for early accident anticipation over video-derived features, under the assumptions of temporal granularity, first-order Markov dependence, and time-invariant transitions.

4.3 Research Questions and Experiments

In this section we dive into the research questions that build our methodology and how we will evaluate these. The aim is to build a probabilistic framework that estimates accident risk over time from visual data, with an emphasis on maintaining interpretability. Our experiments are driven by exploring and evaluating the three research questions below:

How accurately and efficiently can objects be detected and tracked in real time using egocentric dash-cam video under diverse real-world driving conditions to support probabilistic accident anticipation?

To address this research question, we develop a deep learning pipeline for real-time object detection and tracking from dash-cam video sequences. We adopt YOLOv11 for detection, chosen for its improved accuracy on small objects and computational efficiency, and DeepSORT for maintaining object identities across frames in dynamic traffic scenes. To ensure trajectories reflect true object motion rather than camera-induced effects, we apply ego-motion compensation using optical flow–based visual odometry Muller and Savakis [2017]. This stabilization enables reliable extraction of motion features such as velocity, proximity, and lane position, forming the structured inputs required for downstream probabilistic reasoning about accident risk.

We evaluate this research question by using tracking metrics like mean average precision (mAP), Intersection over union (IOU), Multiple Object Tracking Accuracy (MOTA) and ID switch count. We evaluate feature reliability by measuring the temporal consistency and completeness rates across frame sequences.

To what extent can structure learning methods recover the spatio-temporal dependencies among observed, latent risk states and other latent factors from ego-centric visual data, and how does the learned structure balance predictive accuracy and computational feasibility for real-time accident anticipation?

To address this research question, we employ a Dynamic Bayesian Network(DBN) with hierarchical latent structures that represents accident risk through multiple levels of latent abstraction. Rather than modelling accident risk as directly dependent on observable features, we introduce intermediate latent factors at a middle layer that capture distinct aspects of driving risk: Collision Imminence (capturing temporal proximity dynamics from time to accident(TTA) and closing speeds), Behavioural Risk (inferred from lane deviations, acceleration patterns, driver intent), and Environmental Hazard (weather, lighting, and road surface indicators).These intermediate latent variables are themselves represented as discrete hidden states that evolve temporally via first-order Markov transitions, and collectively influence the top-level latent Accident Risk state.

This hierarchical structuring of latent variables constitutes our primary contribution, rather than a direct observation-to-risk mapping, we explicitly model the multi-level dependency pathway through which observable features manifest as intermediate risk factors that ultimately determine overall accident risk. We apply the DYNOTEARS framework Pamfil *et al.* [2020] to learn structure of the temporal dependencies within

this hierarchical architecture. DYNOTEARS extends continuous optimization methods to dynamic Bayesian networks, enabling the discovery of acyclic graph structures that capture both temporal and cross-variable relations. Although a DAG will be learnt it is important to note that in this current state we discover correlations in the data and not direct causality. In our context, DYNOTEARS automatically discovers which observable features most strongly influence which intermediate latent factors, whether direct paths exist from observations to accident risk or if all influence is mediated through intermediate factors, how intermediate latent factors interact with each other, and the temporal lag structure showing how quickly risk escalates through the hierarchy. This data-driven structure discovery is combined with domain knowledge to initialize the hierarchy, then refined through Expectation-Maximization for latent variable parameter estimation given the learned structure.

We assess the learned structure through two primary dimensions.First is predictive Accuracy where we evaluate accident anticipation performance using standard metrics (AUC, F1-score, Time-to-Accident at various horizons). We will be evaluating these key performance metrics to see how the learnt structure is able to perform against deep learning architectures.

Second is Computational feasibility. We measure structure learning time on the full dataset, real-time inference speed, memory requirements for storing learned parameters, and scalability analysis showing how computational demands grow with the number of variables and temporal window length.

Given the absence of ground truth causal structures for accident risk hierarchies and the inherent subjectivity in assessing model interpretability, we acknowledge that formal validation of the discovered structure's correctness or interpretability is not feasible.

Given a learned dynamic probabilistic graph structure, how effective are inference methods in real-time accident anticipation, and how do the hierarchical explanations showing risk propagation through intermediate latent factors provide interpretable reasoning that we can validate against domain knowledge ?

To address this research question, we evaluate probabilistic inference methods on the learned hierarchical DBN structure for real-time accident anticipation. Given the observable features extracted from dash-cam video at each time step, we perform approximate inference using Loopy Belief Propagation and variational inference techniques to compute the posterior probability distribution over the latent Accident Risk states and the intermediate latent factors. These approximate inference techniques provide tractable computation while maintaining reasonable accuracy for temporal graphical

models.

We measure predictive effectiveness using area under curve (AUC) for binary accident/no-accident classification at various time horizons, F1-score for multi-class risk state prediction, and Time-to-Accident (TTA) metrics capturing how early the model correctly predicts elevated risk states before collision. We evaluate the predictive effectiveness of our methods benchmarked against the deep learning methods from Yi *et al.* [2023],Bao *et al.* [2020],Karim *et al.* [2022] and Selvaraju *et al.* [2017].

To assess whether our hierarchical explanations provide transparent reasoning, we analyse Risk propagation pathways by tracing how observable features influence intermediate latent factors, which in turn determine top-level accident risk. We also use State transition coherence by examining whether learned temporal transitions follow expected monotonic risk escalation patterns according to expert domain knowledge, ensuring the inferred reasoning is transparent and trustworthy in safety-critical contexts. This validation ensures the learned structure produces explanations that stakeholders can trust and verify against domain expertise, rather than opaque predictions requiring post-hoc rationalization.

4.3.1 Dataset

The Car Crash Dataset (CCD) is a publicly available video dataset designed for training and evaluating models on traffic accident anticipation using ego-centric dash-cam footage. It contains a total of 4,500 short video clips, each lasting approximately 5 seconds and sampled at 10 frames per second (FPS), resulting in 50 frames per video. Of these videos, 1,500 contain actual road accidents, while the remaining 3,000 clips depict normal driving scenarios without any collisions.

Each video is captured from a forward-facing dash camera, simulating the perspective of the driver or an autonomous vehicle.The dataset includes various environmental and traffic conditions, such as different lighting (day/night), weather (clear/rainy), and road structures (highways, intersections, pedestrian zones).

This dataset serves as the benchmark dataset to train and evaluate the proposed accident anticipation framework. The inclusion of both accidental data and normal situations is essential for learning accurate probabilistic reasoning and risk estimation in dynamic scenes.

As an alternative dataset we will use the DAD (Dashcam Accident Dataset) which is based in six major Taiwanese cities. This dataset has 620 positive clips and 1130 negative clips, each having 100 frames over 5 seconds.

4.3.2 semi supervised annotation

In our hierarchical DBN framework, the latent variable representing Accident Risk is modelled as a discrete random variable with three mutually exclusive states: Safe, Elevated and Critical. Given the 5-second duration of our video clips and the absence of explicit frame-level risk annotations, we employ a semi-supervised annotation strategy that combines temporal proximity and feature-based heuristics. For accident sequences, frames are labelled according to their time from the collision event (i.e Safe $\geq 2.5s$ before collision, Elevated is 2.5s to 1.5s and Critical $\leq 1.5s$ before collision), reflecting the principle that risk escalates as collision becomes imminent. This semi-supervised approach provides initial weak labels for model training while allowing the learning algorithm to refine state boundaries through the Expectation-Maximization procedure, which iteratively updates both the latent state assignments and model parameters.

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